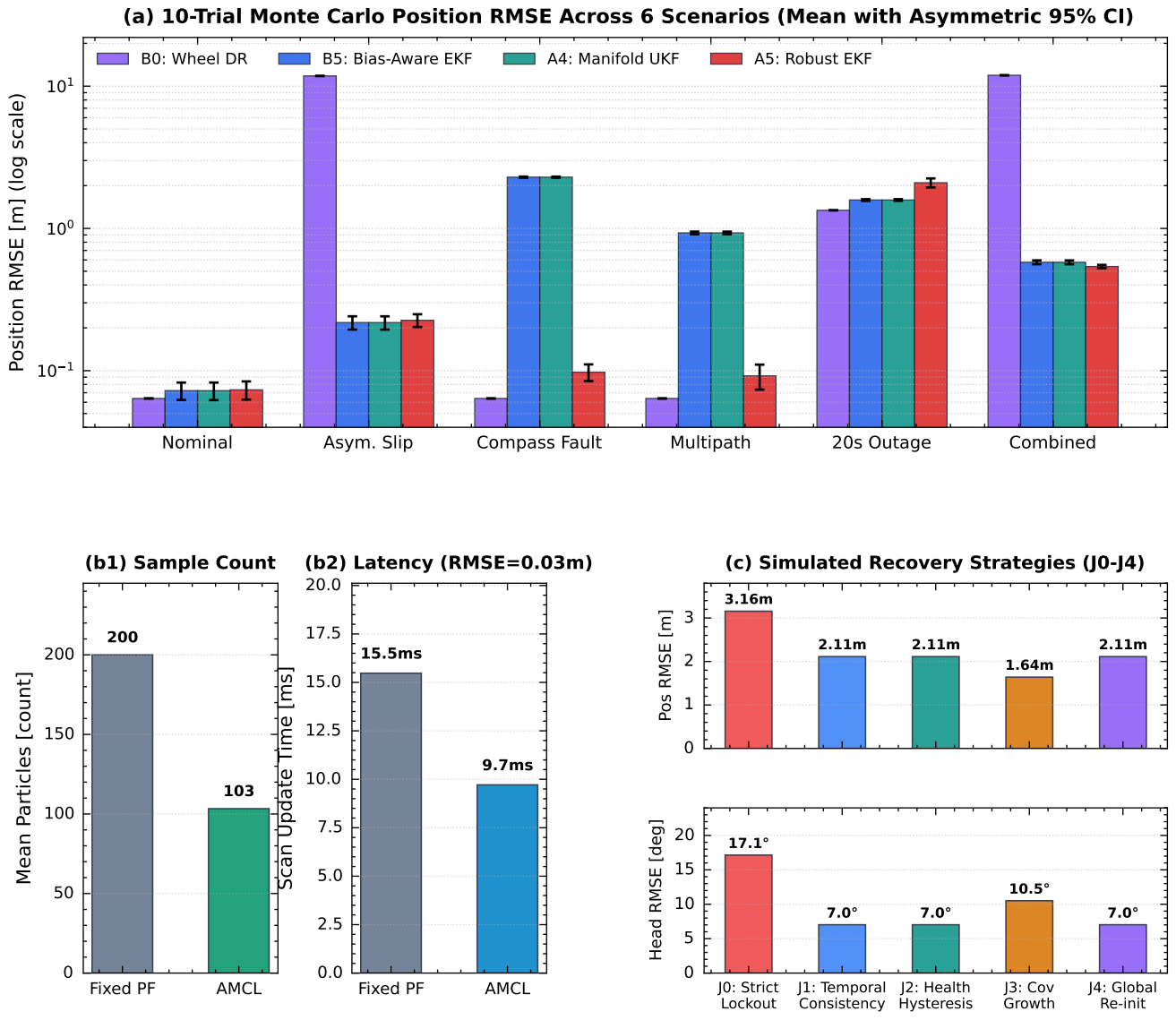
(a) 10-Trial Monte Carlo Position RMSE Across 6 Scenarios (Mean with Asymmetric 95% CI)
B0: Wheel DR
B5: Bias-Aware EKF
A4: Manifold UKF
A5: Robust EKF
Position RMSE [m] (log scale)
Nominal
Asym. Slip
Compass Fault
Multipath
20s Outage
Combined
(b1) Sample Count
Mean Particles [count]
200
103
Fixed PF
AMCL
(b2) Latency (RMSE=0.03m)
Scan Update Time [ms]
15.5ms
9.7ms
Fixed PF
AMCL
(c) Simulated Recovery Strategies (J0-J4)
Pos RMSE [m]
3.16m
2.11m
2.11m
1.64m
2.11m
Head RMSE [deg]
17.1°
7.0°
7.0°
10.5°
7.0°
J0: Strict Lockout
J1: Temporal Consistency
J2: Health Hysteresis
J3: Cov Growth
J4: Global Re-init